

Practice Exam — Solutions

Philosophy 305: Introduction to Formal Methods

2026-04-18

Question 1: Probability and Conditional Probability

Let F = fraudulent, G = flagged.

Given: $\Pr(F) = 0.02$, $\Pr(G \mid F) = 0.90$, $\Pr(G \mid \neg F) = 0.05$.

(a) By the law of total probability:

$$\begin{aligned}\Pr(G) &= \Pr(G \mid F) \Pr(F) + \Pr(G \mid \neg F) \Pr(\neg F) = (0.90)(0.02) + (0.05)(0.98) \\ &= 0.018 + 0.049 = 0.067\end{aligned}$$

Answer: 0.067 (or 6.7%).

(b) By Bayes' theorem:

$$\Pr(F \mid G) = \frac{\Pr(G \mid F) \Pr(F)}{\Pr(G)} = \frac{0.018}{0.067} \approx 0.269$$

Equivalently, build the table:

	Flagged	Not Flagged
Fraud	$0.02 \times 0.90 = 0.018$	$0.02 \times 0.10 = 0.002$
Legitimate	$0.98 \times 0.05 = 0.049$	$0.98 \times 0.95 = 0.931$

$$\Pr(F \mid G) = \frac{0.018}{0.018 + 0.049} = \frac{0.018}{0.067} \approx 0.269$$

Answer: approximately **0.269** (about 26.9%).

(c) Two events are independent if and only if $\Pr(A \cap B) = \Pr(A) \cdot \Pr(B)$, or equivalently, $\Pr(A \mid B) = \Pr(A)$.

We have $\Pr(G \mid F) = 0.90$ but $\Pr(G) = 0.067$. Since $0.90 \neq 0.067$, the events are **not independent**. The probability of being flagged is much higher for fraudulent transactions, which is exactly why the system is useful.

Question 2: Probability Trees and Independence

Let P_1 = pass Part 1, P_2 = pass Part 2.

Given: $\Pr(P_1) = 0.7$, $\Pr(P_2 | P_1) = 0.8$, $\Pr(P_2 | \neg P_1) = 0.3$.

(a) $\Pr(P_1 \wedge P_2) = \Pr(P_1) \times \Pr(P_2 | P_1) = 0.7 \times 0.8 = 0.56$.

Answer: 0.56.

(b) Exactly one pass happens two ways:

- Pass Part 1, fail Part 2: $0.7 \times 0.2 = 0.14$
- Fail Part 1, pass Part 2: $0.3 \times 0.3 = 0.09$

Total: $0.14 + 0.09 = 0.23$.

Answer: 0.23.

Question 3: Expected Utility

(a) Expected monetary value (profits in thousands):

$$\text{EMV}(X) = 0.3(-90) + 0.4(80) + 0.3(100) = -27 + 32 + 30 = 35$$

$$\text{EMV}(Y) = 0.3(10) + 0.4(30) + 0.3(50) = 3 + 12 + 15 = 30$$

Answer: Plan X has higher expected monetary value (\$35,000 vs. \$30,000).

(b) Compare cell by cell:

- Downturn: X gives -90 , Y gives 10 . Y is better.
- Stable: X gives 80 , Y gives 30 . X is better.

Each plan beats the other in at least one state, so **neither plan dominates the other** (not strictly, not weakly).

(c) Total wealth (in thousands) = $100 + \text{profit}$. Utility = $\sqrt{\text{wealth in thousands}}$.

Plan X:

- Downturn: wealth = 10 , utility = $\sqrt{10} \approx 3.162$
- Stable: wealth = 180 , utility = $\sqrt{180} \approx 13.416$
- Growth: wealth = 200 , utility = $\sqrt{200} \approx 14.142$

$$EU(X) = 0.3(3.162) + 0.4(13.416) + 0.3(14.142) = 0.949 + 5.367 + 4.243 = 10.558$$

Plan Y:

- Downturn: wealth = 110 , utility = $\sqrt{110} \approx 10.488$
- Stable: wealth = 130 , utility = $\sqrt{130} \approx 11.402$
- Growth: wealth = 150 , utility = $\sqrt{150} \approx 12.247$

$$EU(Y) = 0.3(10.488) + 0.4(11.402) + 0.3(12.247) = 3.146 + 4.561 + 3.674 = 11.381$$

Answer: The owner should choose Plan Y according to expected utility (approximately 11.38 vs. 10.56). This is the opposite of part (a). Plan X has higher expected monetary value, but the concave (square root) utility function penalises the catastrophic Downturn scenario for X, where wealth drops to just \$10,000. The declining marginal utility of money makes the safer plan preferable.

Question 4: Strategic-Form Games

	A	B	C	D
T	(4, 2)	(2, 1)	(3, 3)	(1, 0)
M	(2, 3)	(4, 0)	(1, 2)	(3, 1)
B	(1, 1)	(3, 2)	(2, 4)	(0, 0)

(a) Check Row's strategies pairwise:

- T vs. M: against A, T gives $4 > 2$; against B, T gives $2 < 4$. No dominance.
- T vs. B: against A, T gives $4 > 1$; against B, T gives $2 < 3$. No dominance.
- M vs. B: against A, M gives $2 > 1$; against C, M gives $1 < 2$. No dominance.

No strategy dominates another for Row.

Check Column's strategies pairwise (Column's payoffs):

- A vs. D: against T, A gives $2 > 0$; against M, A gives $3 > 1$; against B, A gives $1 > 0$. **A strictly dominates D.**
- C vs. D: against T, C gives $3 > 0$; against M, C gives $2 > 1$; against B, C gives $4 > 0$. **C strictly dominates D.**
- C vs. B: against T, C gives $3 > 1$; against M, C gives $2 > 0$; against B, C gives $4 > 2$. **C strictly dominates B.**
- No other strict dominance relations hold. (For instance, A vs. B: against B, A gives $1 < 2$, so A does not dominate B.)

For Column, A and C both strictly dominate D, and C strictly dominates B. No dominance relations for Row.

(b) Iterated elimination of strictly dominated strategies:

Round 1: D is strictly dominated (by A, and by C) for Column, and B is strictly dominated by C. Eliminate both D and B.

(Alternatively, eliminate them one at a time in two rounds; the order doesn't matter.)

	A	C
T	(4, 2)	(3, 3)
M	(2, 3)	(1, 2)
B	(1, 1)	(2, 4)

Round 2: Now check Row. T vs. M: T gives $4 > 2$ against A and $3 > 1$ against C. **T strictly dominates M.** T vs. B: T gives $4 > 1$ against A and $3 > 2$ against C. **T strictly dominates B.**

Eliminate M and B.

	A	C
T	(4, 2)	(3, 3)

Round 3: Column chooses between A (payoff 2) and C (payoff 3). Column picks C.

Iterated elimination yields (T, C) with payoffs (3, 3).

(c) Best responses:

Row's best responses:

- Against A: T gives 4, M gives 2, B gives 1. Best: **T**.
- Against B: T gives 2, M gives 4, B gives 3. Best: **M**.
- Against C: T gives 3, M gives 1, B gives 2. Best: **T**.
- Against D: T gives 1, M gives 3, B gives 0. Best: **M**.

Column's best responses:

- Against T: A gives 2, B gives 1, C gives 3, D gives 0. Best: **C**.
- Against M: A gives 3, B gives 0, C gives 2, D gives 1. Best: **A**.
- Against B: A gives 1, B gives 2, C gives 4, D gives 0. Best: **C**.

Check each cell: the only cell where both are best responses is (T, C), since T is Row's best response to C and C is Column's best response to T.

The unique pure-strategy Nash equilibrium is (T, C) with payoffs (3, 3). This is consistent with the iterated elimination result.

Question 5: Dynamic Games and Backward Induction

(a) Player 1 moves at two nodes: the root (A or B) and after $B \rightarrow F$ (G or H). So Player 1 has $2 \times 2 = 4$ strategies:

1. (A; G) — play A at root, play G if the game reaches the second node.
2. (A; H) — play A at root, play H if the game reaches the second node.
3. (B; G) — play B at root, play G if the game reaches the second node.
4. (B; H) — play B at root, play H if the game reaches the second node.

Player 2 moves at two nodes: after A (C or D) and after B (E or F). So Player 2 has $2 \times 2 = 4$ strategies:

1. (C; E) — play C after A, play E after B.
2. (C; F) — play C after A, play F after B.
3. (D; E) — play D after A, play E after B.
4. (D; F) — play D after A, play F after B.

Player 1 has 4 strategies. Player 2 has 4 strategies.

(b) Work backwards:

Step 1 — Player 1's second node (after B, F): G gives Player 1 a payoff of 6, H gives 1. Player 1 chooses **G**. Replace this node with (6, 0).

Step 2 — Player 2's node after B: E gives Player 2 a payoff of 4, F leads to (6, 0) so Player 2 gets 0. Player 2 chooses **E**. Replace with (3, 4).

Step 3 — Player 2's node after A: C gives Player 2 a payoff of 3, D gives 1. Player 2 chooses **C**. Replace with (5, 3).

Step 4 — Player 1's root node: A leads to payoff 5, B leads to payoff 3. Player 1 chooses **A**.

Outcome: $A \rightarrow C$, with payoffs (5, 3).

(c) Player 2 prefers (3, 4) from $B \rightarrow E$. To get Player 1 to choose B, Player 2 could threaten: "If you play A, I will play D, giving you only 2."

If Player 1 believed this, they would compare $A \rightarrow D$ (payoff 2) with $B \rightarrow E$ (payoff 3) and choose B.

But the threat is **not credible**. At Player 2's decision node after A, C gives Player 2 a payoff of 3 and D gives Player 2 a payoff of 1. A rational Player 2 will play C. Player 1 can anticipate this, so the threat doesn't change Player 1's reasoning.

(d) The strategic form table (using the strategies listed in part (a)):

	(C; E)	(C; F)	(D; E)	(D; F)
(A; G)	(5, 3)	(5, 3)	(2, 1)	(2, 1)
(A; H)	(5, 3)	(5, 3)	(2, 1)	(2, 1)
(B; G)	(3, 4)	(6, 0)	(3, 4)	(6, 0)
(B; H)	(3, 4)	(1, 5)	(3, 4)	(1, 5)

Notice that the rows for (A; G) and (A; H) are identical. When Player 1 chooses A, the G/H choice is never reached, so it doesn't affect the outcome. The difference between these strategies only matters in the context of backward induction and subgame perfection.

Question 6: Mixed Strategies

	Left	Right
Up	(3, 1)	(1, 2)
Down	(1, 3)	(2, 0)

Let Row play Up with probability p and Down with probability $1 - p$.

Set Column indifferent between Left and Right:

Column's expected payoff from Left: $1 \cdot p + 3 \cdot (1 - p) = 3 - 2p$

Column's expected payoff from Right: $2 \cdot p + 0 \cdot (1 - p) = 2p$

Set equal: $3 - 2p = 2p$, so $4p = 3$, giving $p = 3/4$.

Let Column play Left with probability q and Right with probability $1 - q$.

Set Row indifferent between Up and Down:

Row's expected payoff from Up: $3q + 1(1 - q) = 2q + 1$

Row's expected payoff from Down: $1 \cdot q + 2(1 - q) = 2 - q$

Set equal: $2q + 1 = 2 - q$, so $3q = 1$, giving $q = 1/3$.

Mixed-strategy Nash equilibrium: Row plays Up with probability $\frac{3}{4}$ and Down with probability $\frac{1}{4}$. Column plays Left with probability $\frac{1}{3}$ and Right with probability $\frac{2}{3}$.

Expected payoffs: Row's expected payoff is $2 \times \frac{1}{3} + 1 = \frac{5}{3} \approx 1.67$. Column's expected payoff is $3 - 2 \times \frac{3}{4} = \frac{3}{2} = 1.5$.

END OF SOLUTIONS